

Lecture 4: Wave reflection and transmission at planar interfaces — Part 1. Normal incidence

Physical principles of remote sensing
(Ge 158)
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Remote sensing image of the day



Sun dog over Antarctica (credit: British Antarctic Survey)

Topics covered

- Reflection and transmission of waves at planar interfaces: Special case of normal incidence
 - Transverse waves on a string (intuition building)
 - EM boundary conditions
 - EM waves w/ normal incidence to a plane
- (Next few classes: Oblique incidence to planar surfaces, multiple surfaces, multiple layers, and introduction to scattering from non-planar surfaces)

Summary

- Normal incidence to a planar boundary
 - Angular frequency is constant across the boundary (necessary for continuity)
 - Reflection and transmission are a functions of the relative phase velocities (eq., wavenumbers and wavelengths) in the media; true for all wave types, not just EM waves
 - For EM waves, reflection and transmission depend on square roots of dielectric constants (i.e., indices of refraction)
- Boundary conditions (BCs) for EM waves can be derived from Maxwell's Equations. For RS applications,
 - Component of electric displacement (ϵE) normal to interface is conserved
 - Component of H field normal to interface is conserved
 - Components of **E** field tangent to the interface are conserved
 - Components of **H** field tangent to the interface are conserved
- Applying BCs to EM wave gives Fresnel Equations for normal incidence

$$\rho = \frac{E_0^r}{E_0^i} = \frac{\eta_2 - \eta_1}{\eta_2 + \eta_1} = \frac{\sqrt{\epsilon_1} - \sqrt{\epsilon_2}}{\sqrt{\epsilon_1} + \sqrt{\epsilon_2}} \quad \tau = \frac{E_0^t}{E_0^i} = \frac{2\eta_2}{\eta_2 + \eta_1} = \frac{2\sqrt{\epsilon_1}}{\sqrt{\epsilon_1} + \sqrt{\epsilon_2}}$$

- When $\epsilon_2 < \epsilon_1$, reflected wave is in phase; else out of phase (i.e., phase shift by π)

Remote sensing applications

- Ice (or ground) penetrating radar
- Example of a multi-layer, normally incident EM wave
- Used to locate and characterize glacier beds and internal layers (esp. volcanic ash deposits); ice thickness, subglacial hydrology, erosion/sedimentation, context for ice cores, paleo-ice-flow studies...
- Active areas of research (in glaciology): Quantifying liquid water content at glacier beds, observing spatiotemporal changes in subglacial hydrology, estimating englacial temperatures and liquid water content

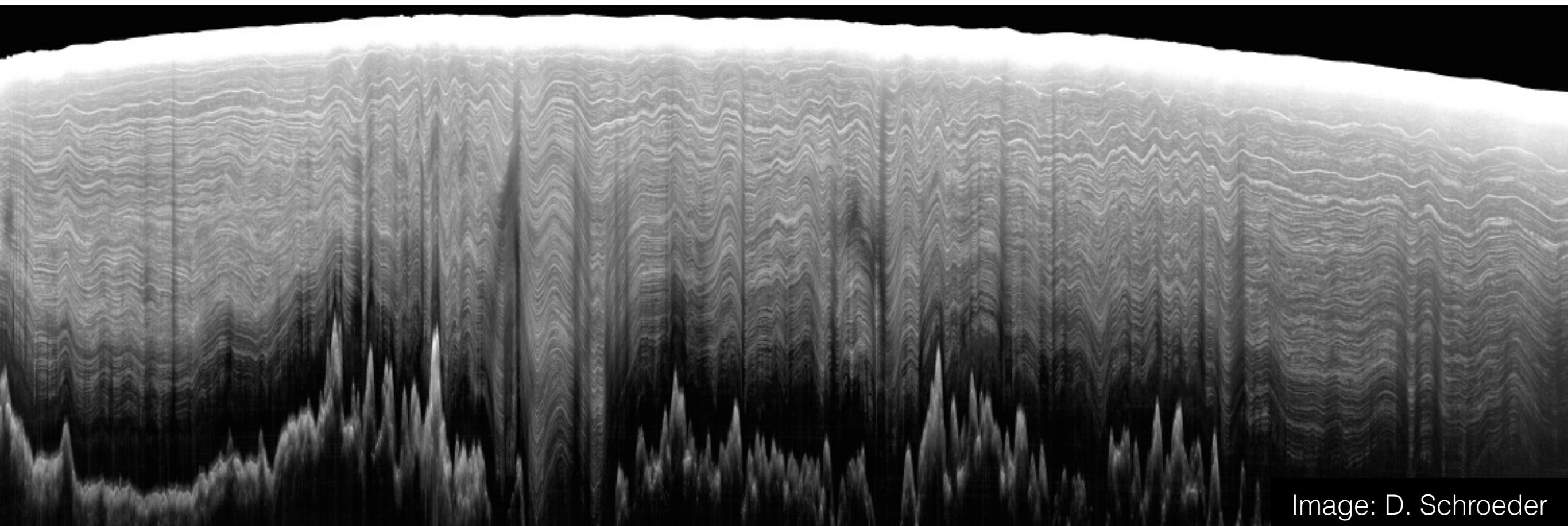


Image: D. Schroeder